these areas suggesting that the speed limits have been effective (Laist et al., 2014).

3.16 Several models have shown that speeds between 10 and 13 knots drastically decrease the probability of lethal injuries in case of collisions between ships and cetaceans (Vanderlaan & Taggart 2007; Gende et al., 2011; Conn & Siliber 2013). There is strong support to identify 12 knots (11.8 knots or 6.1 m/s) as Bayesian change point of probability for the relationship between ship speed and encounter distance. Average encounter distances above and below the 11.8 knots change point vary from 448 m (95%CrI, 398-485) to 562 m (95%CrI, 468-676) (Gende et al., 2011).

Physical disturbance of cetaceans by ships

3.17 The presence of ships may influence cetaceans: attraction, flight or no apparent reaction, depending on species and individuals (Di-Méglio et al., 2010). It is likely to generate behavioural responses causing individuals to move to less favourable hab itats, altering the

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SOURCE: MEPC.380(80).pdf
reaction, depending on species and individuals (Di-Méglio et al., 2010). It is likely to generate behavioural responses causing individuals to move to less favourable hab itats, altering the normal course of functions such as foraging, social functioning, reproduction, suckling, resting or migration. This state of stress alters the health status of individuals and demographic parameters may be degraded. If changes in cetacean behaviour have been observed (notably in the case of the bottlenose dolphin in the Mediterranean) and disturbance distances have sometimes been inferred, it is difficult in the current state of knowledge to quantify the impacts of this pressure in terms of population ecology.

Underwater noise from commercial shipping

3.18 Underwater noise generated by human activities is one of the pressures identified and assessed in the framework of the implementation of the Marine Strategy Framework Directive (descriptor 11 of the Directive) and its complementary process at the Medi terranean level (Ecosystem Approach Process (EcAp) led by the Barcelona Convention). Among the activities

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(descriptor 11 of the Directive) and its complementary process at the Medi terranean level (Ecosystem Approach Process (EcAp) led by the Barcelona Convention). Among the activities concerned is shipping, where the main contributor to the noise generated by a merchant ship is the movement of the engine propeller. The noise level i ncreases with the shape of the propeller, the state of wear of the ship, its size, speed and loading. The literature shows a direct relationship between speed and noise (McKenna et al., 2013; Zobell et al., 2021). Leaper (2019) concluded that a 10% speed reduction would reduce the total sound energy from shipping by around 40% on the global scale.

3.19 In the Mediterranean basin, anthropogenic noise levels have been steadily increasing over the past 50 years as shipping traffic has increased. According to the first EU maritime transport first environmental impact report (EMTER report) published in 2 021, for EU waters the total accumulated underwater radiated noise energy more than doubled between 2014 and 2019. The underwater radiated noise (URN) from shipping is now recognized both in IMO

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the total accumulated underwater radiated noise energy more than doubled between 2014 and 2019. The underwater radiated noise (URN) from shipping is now recognized both in IMO and EU as a significant environmental issue with regional and global impact. The European Maritime Safety Agency (EMSA) conducted a study in 2021, focusing on a number of key aspects related to URN: the existing policy and current understanding about sources of continuous URN from different types of ships, its impacts on the marine environment, and mitigation actions. The study was carried out by "WavEC Offshore Renewables" and "Maritime Research Institute Netherlands" (MARIN) on behalf of EMSA. Commercial vessels can have short- and long-term negative consequences for marine life, in particular marine mammals (IMO, 2014, MEPC.1/Circ.833): the diffuse increase by maritime traffic in ambient noise levels, especially in the low frequencies, reduces the communication range of cetaceans, making it difficult for them to find mates or establish social relationships, as well as foraging and orientation. Furthermore, repeated shallow dives to cope with persistent acoustic disturbance

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difficult for them to find mates or establish social relationships, as well as foraging and orientation. Furthermore, repeated shallow dives to cope with persistent acoustic disturbance are likely to increase the risk of decompression illness in marine mammals (GIS3M, 2010).
RESOLUTION MEPC.380(80) (adopted on 7 July 2023)
DESIGNATION OF THE NORTH-WESTERN MEDITERRANEAN SEA AS A PARTICULARLY SENSITIVE SEA AREAMEPC 80/17/Add.1
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3.20 To be noted, ships concerned with speed reductions should be chosen carefully, as these measures can also have opposite effects on underwater noise and gas emissions depending on propeller designs (Leaper, 2019), and the technical criteria of the electrical distribution and the type of propulsion of the ship. As the aim of this project is not to increase the impact of maritime traffic on cetaceans, consideration should be given to the equipment of vessels to reduce noise. For example, changing the propellers during maintenance, having a certificate of conformity, equipping with a noise self-estimation and cavitation detection system.

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SOURCE: MEPC.380(80).pdf
vessels to reduce noise. For example, changing the propellers during maintenance, having a certificate of conformity, equipping with a noise self-estimation and cavitation detection system. The designation of the PSSA will allow further studies to be carried out on the matter.

Chemical pollution

Hydrocarbons

3.21 Accidental oil spills have become rare in the Mediterranean, the last major accident being the MT Haven in the Gulf of Genoa in 1991, but they can cause considerable damage to the marine environment given the quantities of oil spilled and the length o f time it takes for the impacted habitats to recover.

3.22 With regard to illegal discharges, the use of satellite imagery can contribute to the estimation of the number of oil spills from ships, without providing proof that the discharge is

illegal or that it is from a ship. In 2016, EMSA's CleanSeaNet platform recorded a total of 1,073 detections of likely polluting incidents and a total of 1,060 detections of potentially polluting incidents in the Mediterranean region and off the Atlantic coasts of Morocco, Portugal, Spain

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SOURCE: MEPC.380(80).pdf
detections of likely polluting incidents and a total of 1,060 detections of potentially polluting incidents in the Mediterranean region and off the Atlantic coasts of Morocco, Portugal, Spain and France. Although these data re main to be confirmed, both in terms of the nature of the pollution and its origin, they clearly indicate that oil pollution incidents caused by ships remain a concern in the Mediterranean.

3.23 Polycyclic aromatic hydrocarbons (PAHs) can bioaccumulate in the tissues of marine mammals. The viscous crude oil spilled during an oil spill can cover the surface of the cetacean's body for a long period of time, which can reduce its filtering capaci ty: this can be the case for fin whales. The deterioration of zooplankton by an oil spill can also generate an indirect negative impact on some whales, as it is the main food for them.

Anti-fouling paints

3.24 Anti-fouling paints are one of the sources of heavy metals and biocides in Mediterranean waters, particularly off the coast of port areas. Through bioaccumulation, marine mammals can be sensitive to this type of pollution, which can disrupt their immune system and even lead to death.

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SOURCE: MEPC.380(80).pdf
Mediterranean waters, particularly off the coast of port areas. Through bioaccumulation, marine mammals can be sensitive to this type of pollution, which can disrupt their immune system and even lead to death.

Other toxic products

3.25 In addition to oil, hazardous and noxious substances (HNS) accidentally spilled into the marine environment can threaten marine species such as cetaceans. HNS include bulk liquid cargoes (petrochemicals, solvents and liquefied gases, etc.), bulk solid cargoes (fertilizers, etc.) and packaged chemicals. The quantities of HNS accidentally spilled have decreased considerably between 1994 and 2013 in the Mediterranean. Since 2003, the discharge of HNS has become insignificant compared to the period from 1994 to 2002.

Marine litter

3.26 The Mediterranean Sea is one of the most affected areas by marine litter in the world and by plastic in particular, which can constitute up to 90% of the seabed litter. A study by Arcangeli et al. 2020 showed that there is a gradient x10 in density of marine litter from

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and by plastic in particular, which can constitute up to 90% of the seabed litter. A study by Arcangeli et al. 2020 showed that there is a gradient x10 in density of marine litter from

offshore, to coastal and to river. Meaning that marine litters are coming from land through rivers (highest densities) and then are spread over the vast oceanic surface. Its origin is mainly land-based, but it is estimated that ships are the source of almo st a quarter of this litter (Koutsodendris et al., 2008; Loakeimidis et al., 2014).

3.27 Accumulation rates vary greatly and are influenced by several factors, such as the proximity of large cities, coastal artificialization and frequentation, hydrodynamics and maritime activities. The semi -enclosed nature of the Mediterranean basin also explains the high accumulation rates observed. The analysis of this waste shows a great variability in its nature

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activities. The semi -enclosed nature of the Mediterranean basin also explains the high accumulation rates observed. The analysis of this waste shows a great variability in its nature and origin, with the highest quantities located mainly near large cities, river mouths and coastal canyons where currents are slower and strong sedimentation occurs.